

Project Documentation

PET SNES Gamepad Adapter

Project number: 221

Revision: 0

Date: 04.01.2025



SNES Gamepad Adapter for Commodore PET Rev.0

Module Description

Introduction

The SNES Gamepad Adapter for PET works with the respective games found on Bo Zimmerman's website zimmers.net and PETSCII robots by the 8bit guy.

The **source of information** about the hardware requirements is the file `snes_adapter.zip` from zimmers.net. A further source was <https://wiki.superfamicom.org/schematics-ports-and-pinouts#>. The `zimmers.net` file contains references to `texelex.com` and Steve Lewis plus source code in C and BASIC for polling the gamepad.

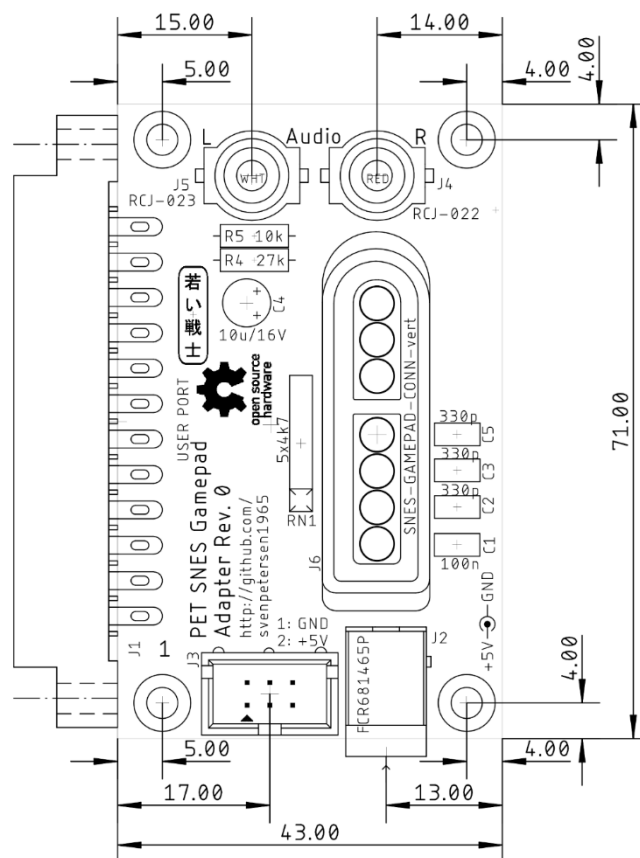


Figure 1: PET SNES Gamepad Adapter Rev. 0 (Dimensions)

Since the PET User Port does not provide +5V supply voltage, there is a 6 was box pin header form connecting a [cassette port dongle](#) (where +5V can be tapped for the Gamepad). This is the recommended solution. Further, there is a 5.5mm/2.1mm barrel connector for a +5V external PSU.

The CB2 audio is tapped from the user port and connected to two RCA jacks via a voltage divider and a capacitor for connecting external PC speakers.

The adapter works with PETSCII Robots for the C64, as well. Since the 5V of the C64 are not connected (that would be the TVVIDEO pin of the user port), either a Cassette Port Dongle or an external Power Supply is required.

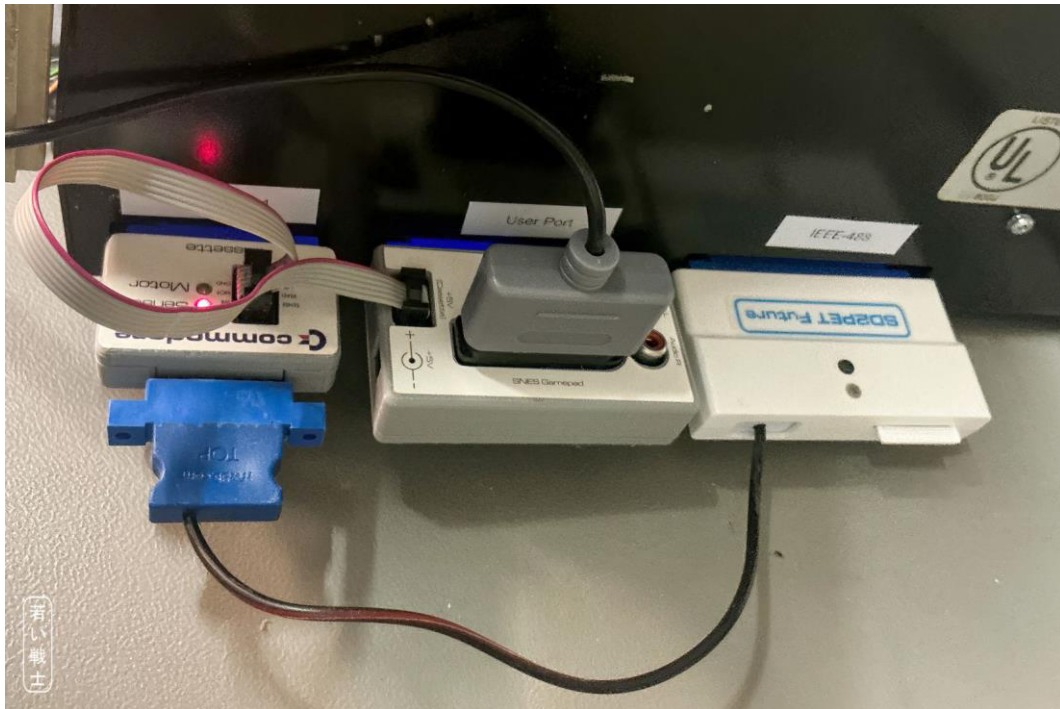


Figure 2: Backside of a CBM8032 - SNES Adapter with Cassette Port Dongle (and SD2PET Future)

Pinouts

J1 - User Port

2x12 Edge Connector, 3.96mm pitch

Pin	Signal	Gamepad
1, 12, A, N	GND	GND
F	PA3	Clock
J	PA5	Latch
K	PA6	Data

J2 – Power Supply Connector

5,5mm/2,5mm barrel connector (Cliff FCR681465P)

Pin	Signal
Center	+5VDC
Shaft	GND

This is an alternative power connector, which can be used with an external +5V power supply. It is **not the preferred** way of powering the module.

J3 - Power connector (from cassette port)

2x3 pin box connector

Signal	Pin	Pin	Signal
GND	1	2	+5V
-	3	4	-
-	5	6	-

Powering from cassette port is the preferred way, since it switches off together with the PET. This requires a Cassette Port Dongle (see [C64 Diagnostic harness](#)).

J4, J5 – Audio RCA jacks

2 vertical RCA jacks

Connector	Color	Center	Ring	Type
J4	Red	Audio	GND	CUI Inc. RCJ-022
J5	White	Audio	GND	CUI Inc. RCJ-023

Note: The CUI Inc connectors are pretty wide spread and available from Mouser and Digikey etc. There are acceptable and cheaper connectors from **AliExpress**, which perfectly fit into the footprint. Click [URL](#)

Both connectors are connected to the same source. The audio level is about 1.23V~ (+4dBu).

J6 – SNES Gamepad

SNES Gamepad Connector (vertical)

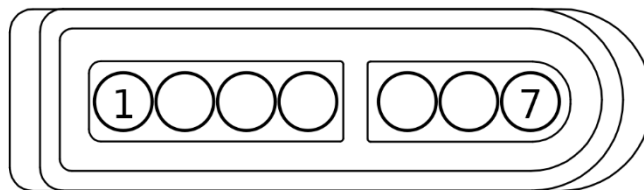


Figure 3: SNES Gamepad Connector

Pin	Signal
1	+5V
2	Clock
3	Latch
4	Data1
5	Data2 (not connected)
6	IOBIT (not connected)
7	GND

Case

There is a 3D printed case and a case label for this project.



Figure 4: PET SNES Gamepad Adapter (in case)

Programming Hints

First, the data direction register of the user port needs to be set correctly. Clock (bit #3) and Latch (bit#5) are outputs of the user port, Data (bit#6) is an input:

```
POKE 59459, 8+32
```

While scanning the gamepad, a (positive) latch pulse is required first:

```
POKE59457, 32:POKE59457, 0:REM LATCH PULSE
```

The gamepad outputs a 12 bit word, MSB first. After the latch pulse, the MSB is present at the data input (bit#6) of the user port.

```
GW=0:B=2048
FOR I=12 TO 1 STEP-1
IF PEEK(59457) AND 64 THEN GW=GW+B:REM IS BIT#6 SET?
B=B/2
POKE59457, 8:POKE59457, 0:REM PULSE CLOCK
NEXT I
```

An active data bit is LOW, this is, why it needs to be inverted

```
GW=4096 + NOT GW:REM INVERT GW
```

Now, the gamepad word can be decoded:

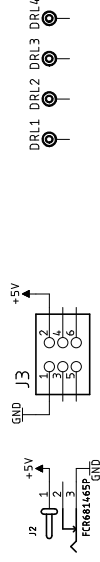
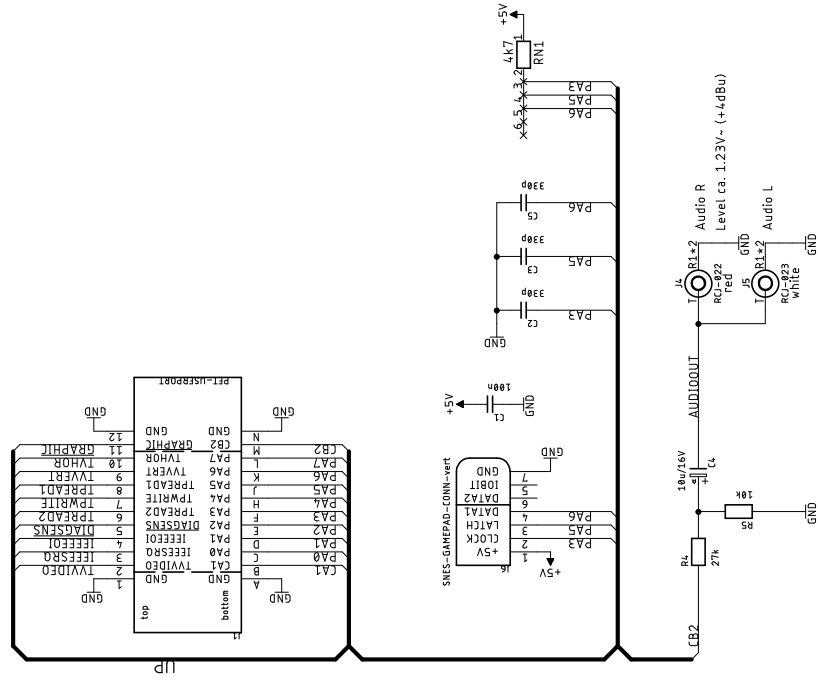
Bit	Bit Value	Gamepad
0	1	Fire-R
1	2	Fire-L
2	4	X
3	8	A
4	16	Right
5	32	Left
6	64	Down
7	128	Up
8	256	Start
9	512	Select
10	1024	Y
11	2048	B

The source is SNES TEST.BAS.PRG from the previously mentioned file snes_adapter.zip

Revision History

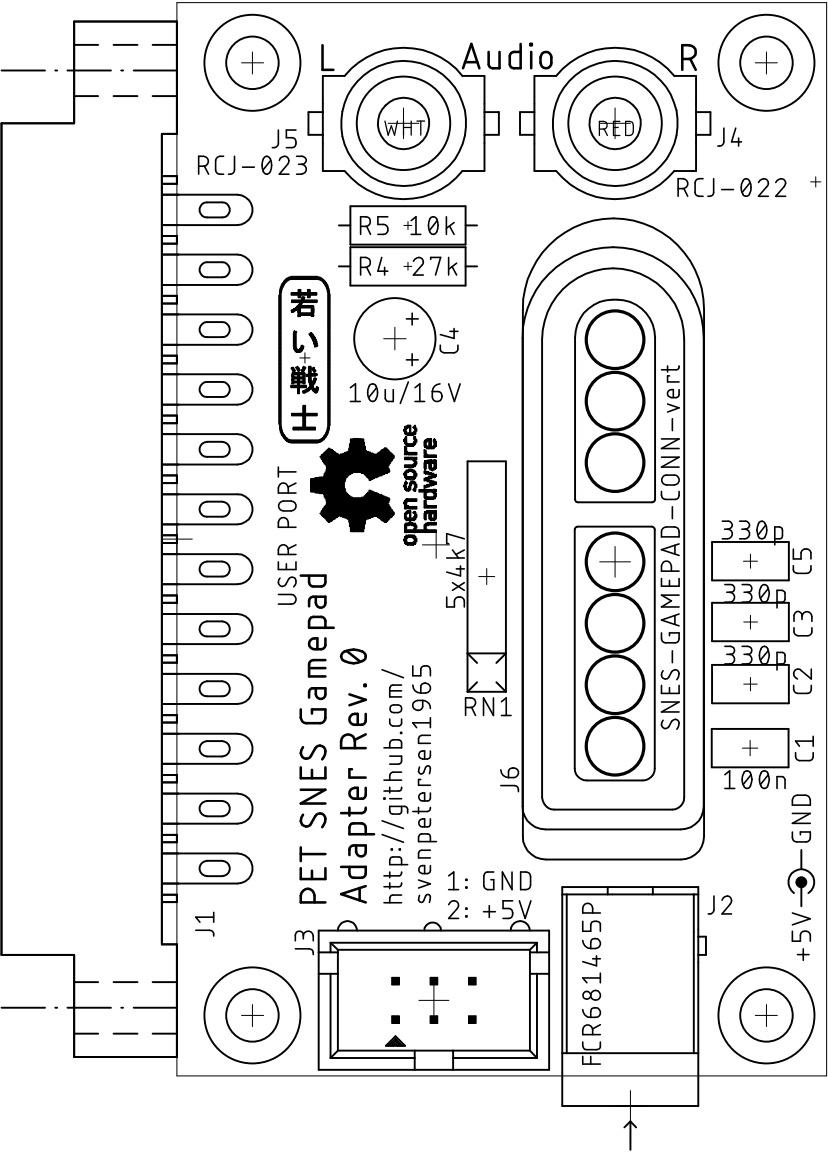
Rev. 0

Prototype, tested ok

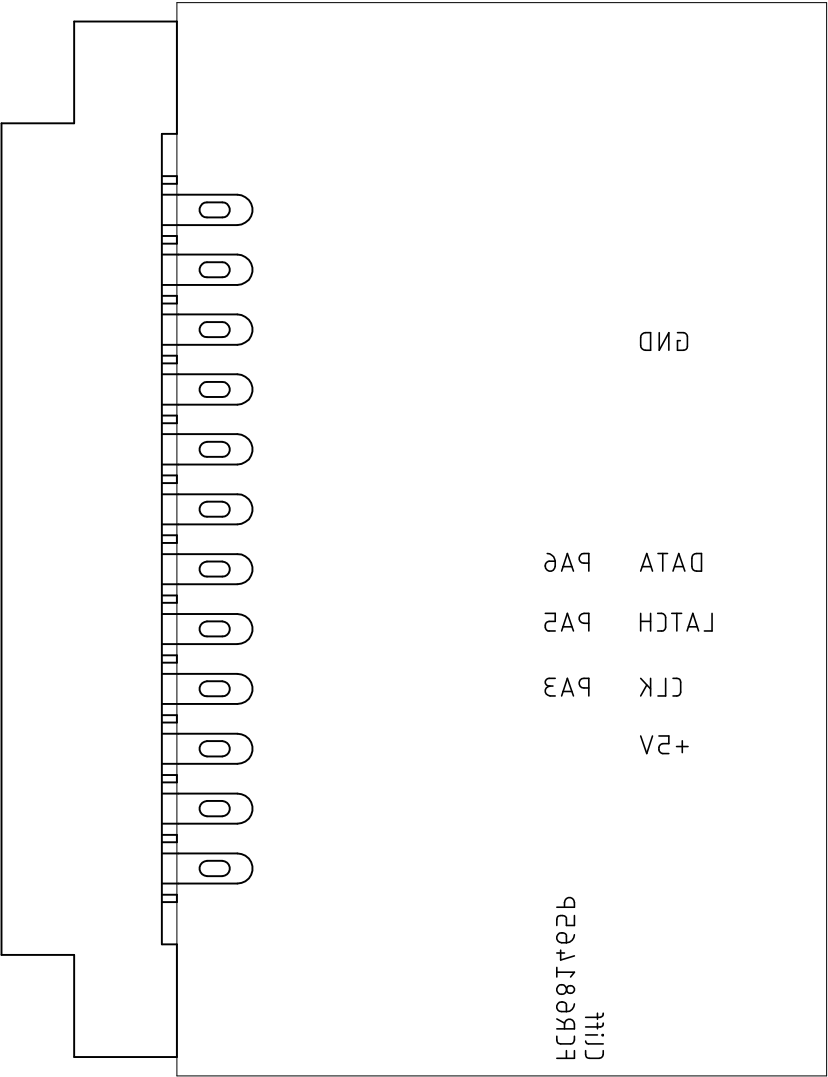


Title:	Commodore PET/CBM SNES Gamepad Adapter			Doc.-No.: 221-1-01-00
Date:	04.01.2025 17:01			Draft: Sven Petersen
File:	PET_snes_gamepad			Rev.: 0 Page 1/1
http://github.com/svenpetersen1965				A3

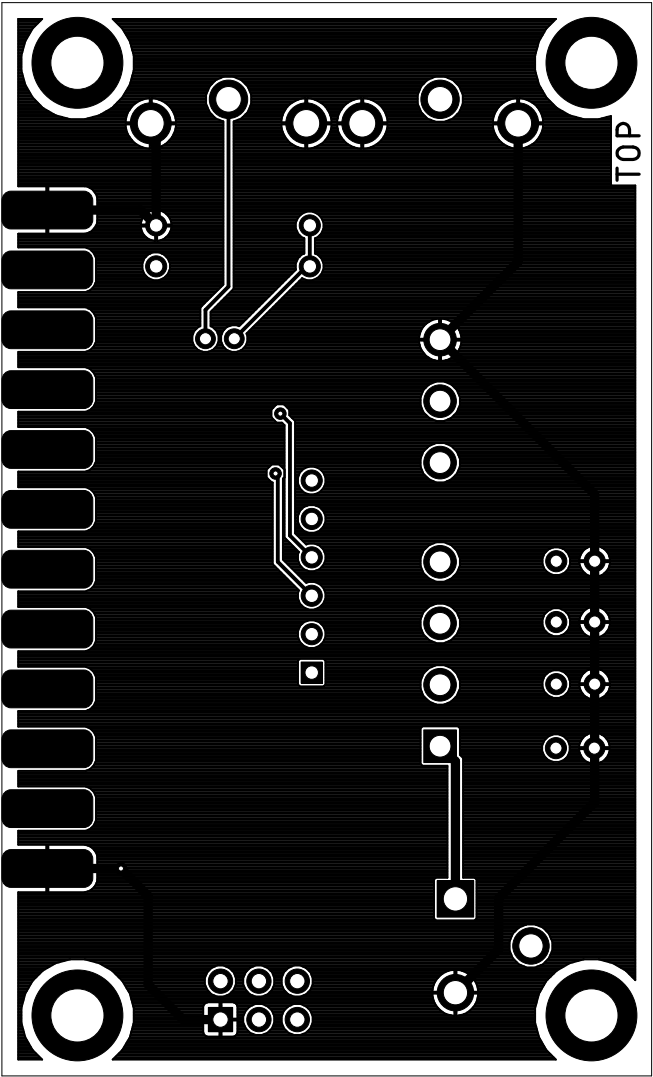
Sven Petersen 2025	Doc.-No.: 221-2-01-00	
	Cu: 35μm	Cu-Layers: 2
PET_snes_gamepad		
09.01.2025 22:25		Rev.: 0
placement component side 9b1z 1əb1o2 1nəməɔɔ61q		



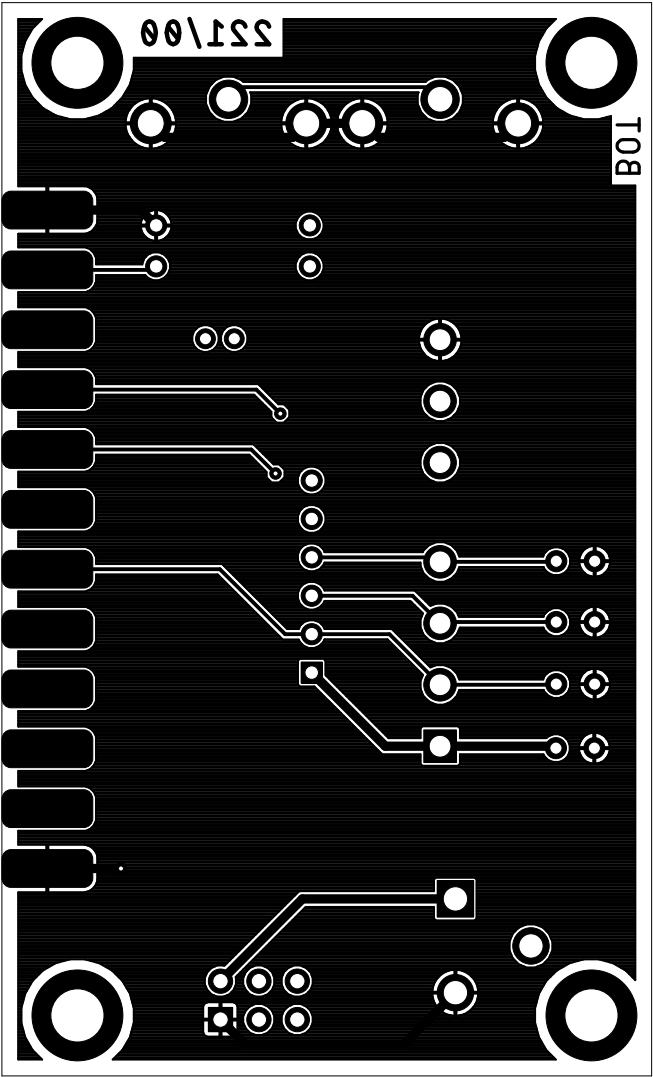
Sven Petersen 2025	Doc.-No.: 221-2-01-00	
	Cu: 35µm	Cu-Layers: 2
PET_snes_gamepad		
09.01.2025 22:25		Rev.: 0
placement order		



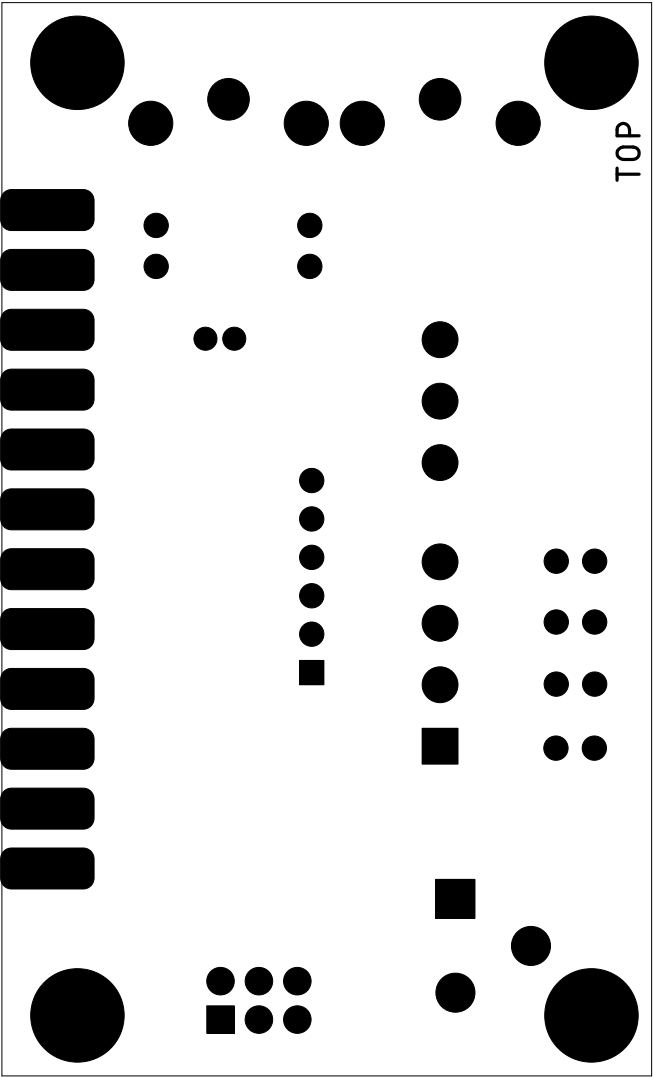
Sven Petersen 2025	Doc.-No.: 221-2-01-00	
	Cu: 35µm	Cu-Layers: 2
PET_snes_gamepad		
09.01.2025 22:25		Rev.: 0
top		



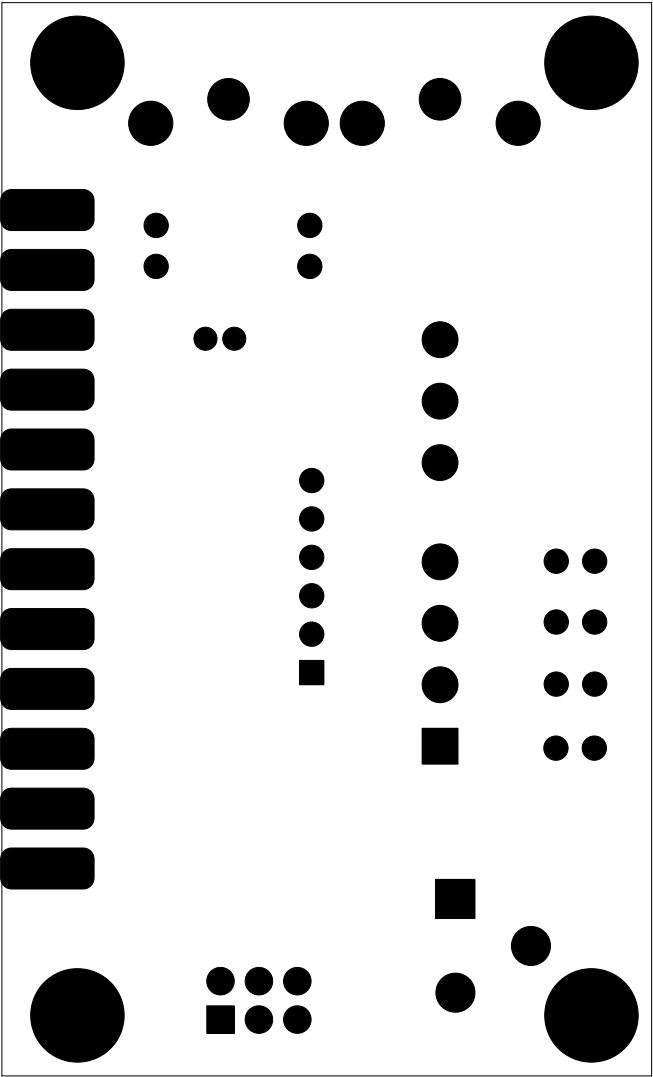
Sven Petersen 2025	Doc.-No.: 221-2-01-00	
	Cu: 35µm	Cu-Layers: 2
PET_snes_gamepad		
09.01.2025 22:25		Rev.: 0
bottom		



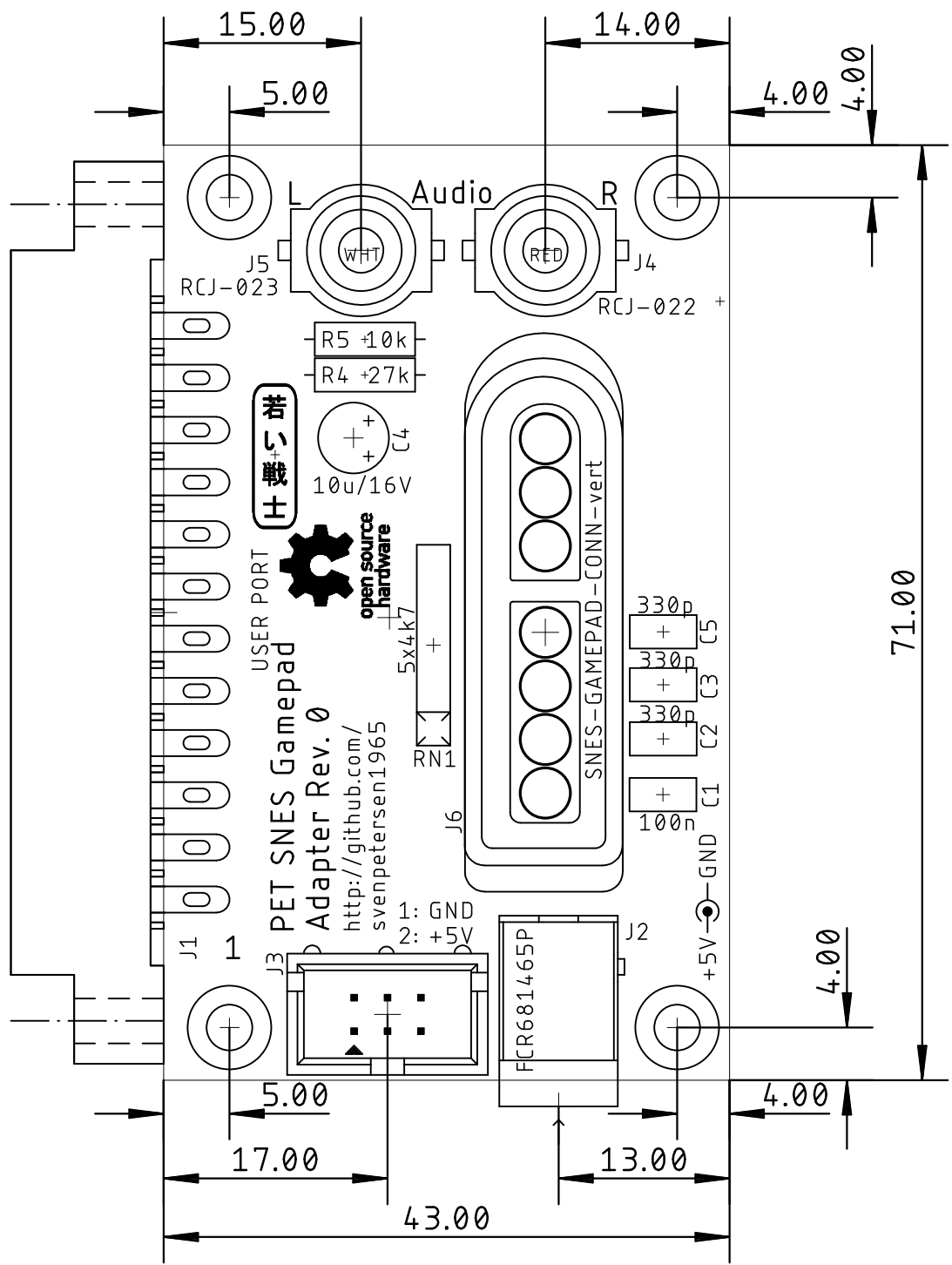
Sven Petersen 2025	Doc.-No.: 221-2-01-00	
	Cu: 35µm	Cu-Layers: 2
PET_snes_gamepad		
09.01.2025 22:25		Rev.: 0
stopmask component side		



Sven Petersen 2025	Doc.-No.: 221-2-01-00	
	Cu: 35µm	Cu-Layers: 2
PET_snes_gamepad		
09.01.2025 22:25		Rev.: 0
stopmask solder side		



Sven Petersen 2025	Doc.-No.: 221-2-01-00	
	Cu: 35µm	Cu-Layers: 2
PET_snes_gamepad		
09.01.2025 22:25		Rev.: 0
placement component side		



SNES Gamepad Adapter for Commodore PET Rev.0

Testing

Test Setup

- PET SNES Gamepad Adapter Rev. 0
- Diagnostic Cassette Port Dongle & 6-way ribbon cable
- A SNES Gamepad (non USB version)
- Commodore CBM3016
- Commodore CBM8032
- Commodore C64
- Software "SNES TEST.BAS.PRG" from zimmers.net (in snes_adapter.zip, see module description)
- Software "PETROBOTS12.D64" (PETSCII Robots paid version from [The 8bit-Guy](#))

The PET SNES Gamepad Adapter was connected to the user port of the respective computer. The Gamepad was connected to the device under test.

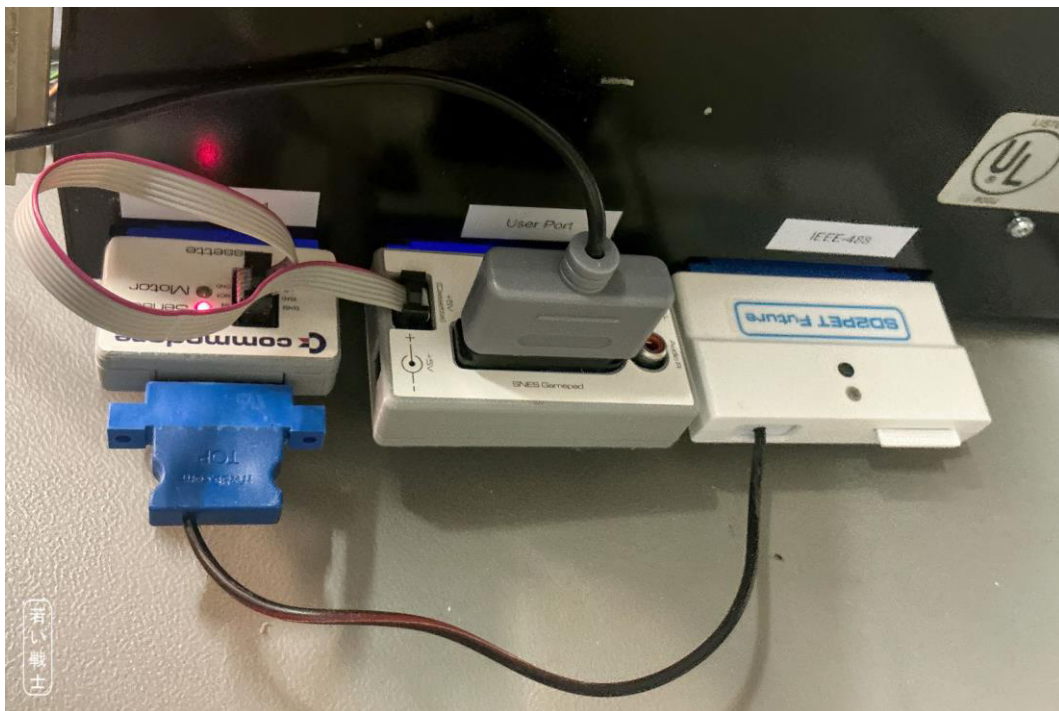


Figure 1: The PET SNES Gamepad Adapter, powered by cassette port and an SD2PET Future with the required software

Test Execution

Testing The Readout of The Gamepad

The computer is the CBM3016, the software is SNES TEST.BAS.PRG. The program was stored on the SD2PET SD card. Loaded and started.

Every button was pressed on the gamepad. The software has display each of that properly.



Figure 2: SNEST TEST.BAS.PRG on CBM3016

The same test was conducted on the CBM8032. Again, every button was displayed properly.



Figure 3: SNEST TEST.BAS.PRG on CBM8032

PETSCII Robots on CBM8032

Due to memory size, the PETSCII Robots software was run on the CBM8032, which had to be set to 40 columns previously.

As a game controller, the SNES Gamepad was selected.



Figure 4: PETSCII Robots on the CBM8032

The software reacted properly to the SNES Gamepad.

PETSCII Robots on C64

The Gamepad was connected to the C64 via the device under test. A cassette dongle was required to power the gamepad. The software was run from the disk image "C64Robots v1.3.d64", which was part of the PETSCII Robots purchase.

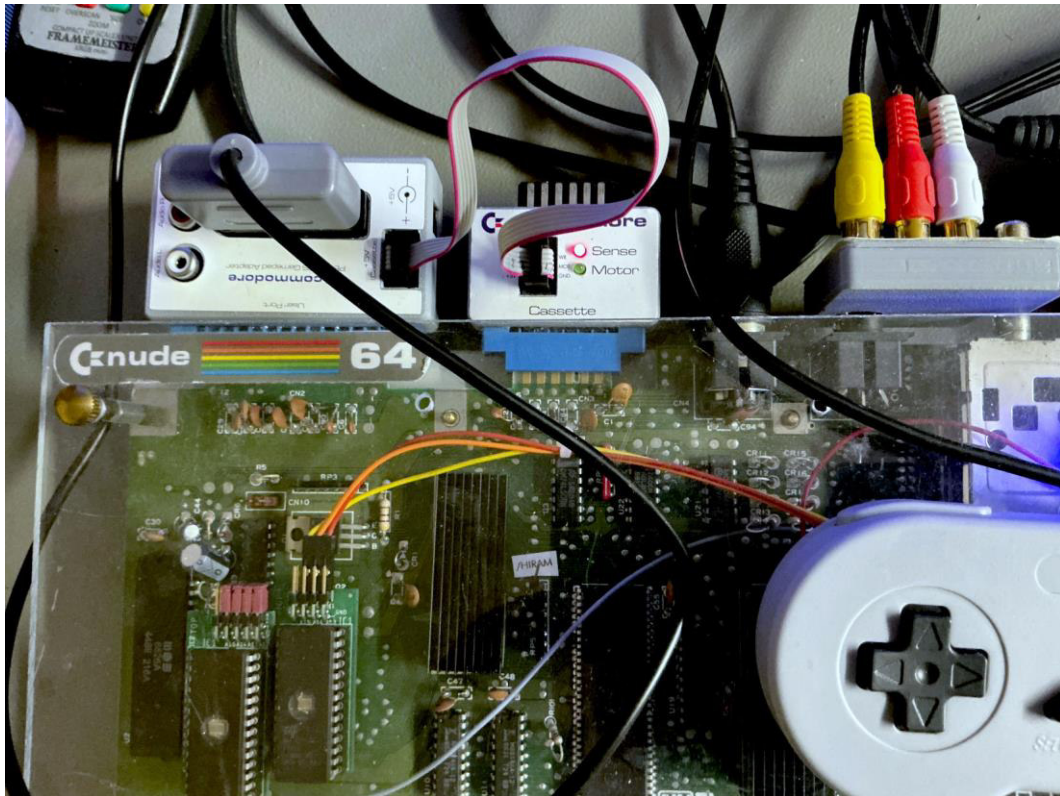


Figure 5: The PET SNES Gamepad Adapter Rev. 0 connected to the C64



Figure 6: PETSCII Robots on the C64

The software reacted properly to the SNES Gamepad.

Conclusion

The PET SNES Gamepad Adapter Rev. 0 is working properly.

PET SNES Gamepad Adapter Rev. 0

Bill of Material Rev. 0.0

Pos.	Qty	Value	Footprint	Ref.-No.	Comment
1	1	221-2-01-00	2 Layer	_PCB Rev. 0	2 layer, Cu 35 μ , HASL, 71.0mm x 43.0mm, 1.6mm FR4
	1	100n	C-2,5	C1	ceramic capacitor, pitch: 2.5mm
	3	330p	C-2,5	C2, C3, C5	ceramic capacitor, pitch: 2.5mm
	1	10u/16V	C5,0/2,0	C4	aluminum electrolytic, Ø5.0mm/pitch 2.0mm 105°C, l=11mm, e.G. Reichelt RD1C106M05011180
	1	Edge connector, 2x12, 3.96mm	USERPORT	J1	edge connector, C64 user port, Ali Express, "series 805" or other
	1	FCR681465P	FCR681465P	J2	Cliff. 2,5mm/5.5mm barrel connector. E.g. Reichelt CLIFF FCR681465P
	1	2x3 box header, 2.54mm	2X03WV	J3	e.g. Reichelt WSL 6G.
	1	RCJ-022	RCJ-02X	J4	CUI Inc., RCA jack, vertical, red. Mouser: 490-RCJ-022, Digikay: CP-1407-ND. Alternative part: Keystone 577
	1	RCJ-023	RCJ-02X	J5	CUI Inc., RCA jack, vertical, white. Mouser: 490-RCJ-023, Digikay: CP-1408-ND. Alternative part: Keystone 584
	1	SNES-GAMEPAD-CONN-vert	SNES_GAMEPAD VERT	J6	Vertical SNES Gamepad Connector from Ali Express
	1	27k	R-10	R4	1/4W, 1% or better
	1	10k	R-10	R5	1/4W, 1% or better
	1	5x4k7	RN-6	RN1	Resistor network 4 pin, 5 resistors